# Reproductive System Embryology

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Block: Endo, Uro, Repro

Thread: Anatomy/Embryology

Conflict of Interest: Dr. Spudich has no Conflicts of interest to disclose

## Textbook Link

Langman's Medical Embryology 15ed: Go to Chapter 16 - Urogenital System
Langman's Medical Embryology 15ed: Go to Chapter 2 - Gametogenesis

## Learning Objectives

By exam time, you should be able to do the following:

1. Describe the germ layer contributions to the development of the organs found in the reproductive systems.
2. Describe the sex hormone driven development of the gonadal and ductal systems.
3. Describe the components of reproductive system development that overlap with the urinary system.
4. Discuss the molecules and signaling pathways needed for normal patterning and development of reproductive systems.
5. Describe structural differences commonly seen across the reproductive systems including driving mechanisms, diagnosis and treatments if known.
6. Compare and contrast gametogenesis in the testes and ovary.

## Study Questions

1. It is said the external genitalia in 46, XX and 46, XY individuals are homologous. What are the anatomic homologues, and what are their embryological origins?

2. After several years of trying to become pregnant, a young patient seeks consultation. Examination reveals she has a bicornate uterus. How might this type of structural difference have occurred?

## A. Genital Ridge/ Indiscriminant phase

1. Genital ridge forms ventral to the mesonephric region and is comprised of intermediate mesoderm. It does not yet contain the primordial germ cells. These are in the yolk sac where they have "lived" since the formation of the epiblast in early development.
   a. Formation of the genital ridge is identical in all individuals

2. Cells on the ridge nearest the abdominal cavity epithelialize, proliferate, and form structures called gonadal cords.
3. Germ cells begin to migrate from the yolk sac to the ridge in week 4 and migration is complete by week 6 when they come to populate the gonadal cords

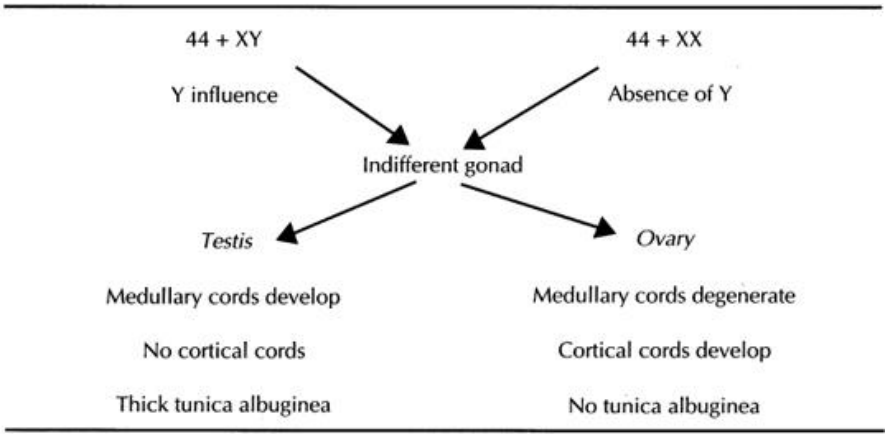
## B. X-Chromosome-Driven Gonadal Differentiation

1. Occurs naturally in the absence of factors from the Y chromosome
2. In the gonad, the gonadal chords form clusters in the cortex, and the medulla becomes a vascular stroma
3. Cord cells envelop the germ cells which have migrated in and will become the follicular cells of the primordial follicle

## C. Y-Chromosome-Driven Gonadal Differentiation

1. Requires the SRY gene from the Y-chromosome which encodes for Testes Determining Factor (TDF)
2. The gonadal cords lengthen and extend deep into the gonad and remain solid until puberty
   a. Epithelial cells of the cords become the Sertoli cells
   b. Mesenchymal cells between the cords become the Leydig cells

3. The germ cells migrate in and stop at the periphery of the cords
4. The entire gonad becomes encased in a capsule called the tunica albuginea




## D. Ductal System – Indiscriminant phase

1. All individuals normally begin with two separate ductal systems.
   a. Mesonephric ducts of the primitive kidneys – called the Wolffian ducts, historically
   b. Paramesonephric ducts – historically referred to as Mullerian ducts

2. Paramesonephric ducts arise from tissue lateral to the genital ridge and have three defined regions
   a. Cranial region is open to the abdominal cavity at the end
   b. Middle region that crosses over the mesonephric ducts towards the midline
   c. Caudal region that fuses with its partner on the opposite side and inserts to the urogenital sinus

## E. Paramesonephric Ductal System Development and Derivatives

1. The uterus, cervix and vagina form from the Paramesonephric system and develop in the presence of estrogen and the absence of Anti-Mullerian Hormone (AMH)
   a. Cranial portions become the fimbriae/infundibulum of the uterine tubes
   b. Middle regions become the bulk of uterine tubes

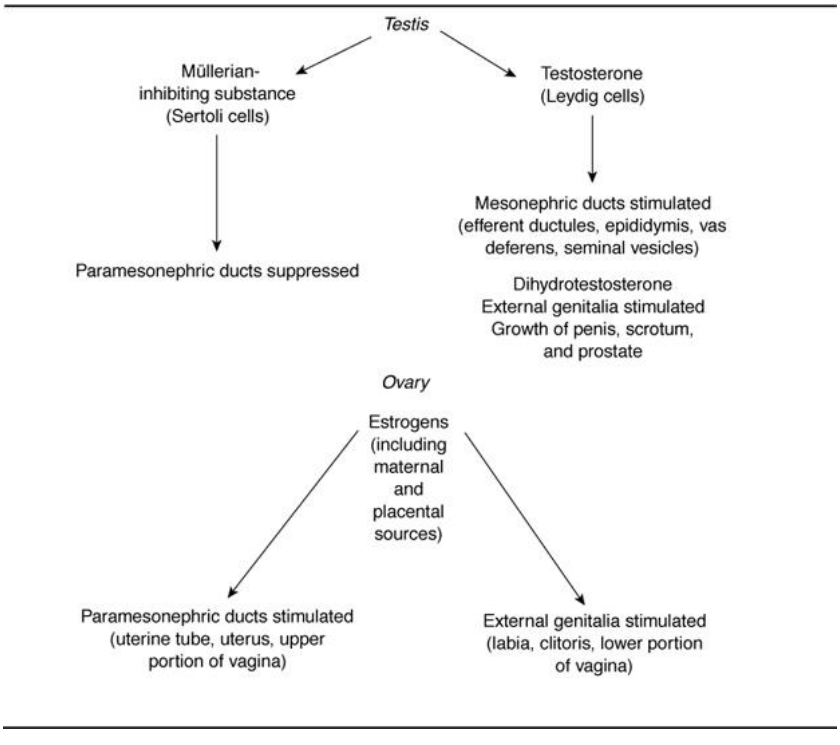
c. Caudal aspects become uterine body, cervix, and upper portions of the vagina

i. Lower portions of vagina from the urogenital sinus

## F. Mesonephric Ductal System Development and Derivatives

1. Comes from the persistent portions of the mesonephric ductal system
   a. Cranial regions become the epididymis and includes some remnants of the OLD mesonephric tubules.
   b. Middle regions form the vas deferens. The seminal vesicles sprout near the caudal end of the vas deferens.
   c. Caudal to the seminal structures is the ejaculatory duct.

2. The Sertoli cells of the newly developing testes secrete AMH in response to testosterone and this drives the degeneration of the paramesonephric structures.



## G. External Genitalia

1. Also goes through an indifferent phase.
2. Cloacal folds form in the perineum are on either side of the cloacal openings
   a. Cranial portions of folds fuse to become the genital tubercle
   b. Urethral folds are in the middle and develop additional genital swellings laterally
   c. Caudal regions fuse as anal folds

3. Testosterone driven differentiation has the following consequences.
   a. Genital tubercle elongates to become the phallus.
   b. Urethral folds wrap around endoderm of cloaca and fuse along ventral side
      i. Cloacal endoderm becomes the penile urethra
   c. Genital swellings become the scrotum.

4. Estrogen driven differentiation has the following consequences
   a. Genital tubercle remains small and develops into the clitoris.
   b. Urethral folds DO NOT fuse but become the labia minora
      i. Urogenital groove persists as the vestibule of the vagina
   c. Genital swellings become the labia majora.

## H. Uterine Structural Differences

1. Uterine abnormalities are many and varied due to abnormal fusion of the distal paramesonephric tubes.
   a. Can cause fertility problems
   b. Some may require surgical repair or assisted reproductive technologies to facilitate pregnancy
   c. Can result from *in utero* exposure to synthetic estrogens like DES

2. Penile spadias

a. Hypospadias where the urethra open along ventral surface, typically underdeveloped. Seen in 3-5 per 1000 births

b. Epispadias are rarer (1 per 30,000) and the urethra is open along the dorsal surface. Often seen in exstrophy of bladders

3. Complete agenesis of external genitalia is rare. Perineum has a urethral opening near the anal opening

4. Differences in the development of external genitalia can occur with either genotype

   a. 46, XX genotype with clitoral hypertrophy

   b. 46, XY genotypes with micropenis

   c. Intersexed individuals with gonads classified as ovotestes are usually 46, XX and may also have a uterus with difficult to classify external genitalia.

5. Androgen Insensitivity Syndromes – are from defective androgen receptor signaling. Affected individuals are 46, XY by genotype but the external genitalia appear phenotypically as 46, XX. They have cryptorchid gonads and no uterus. May be complete, mild, or partial.
6. Congenital Adrenal Hyperplasia causes defective sex differentiation. The most common form when an individual is deficient in 21-hydroxylase with 46, XX individuals being virilized
7. 5-alpha-reductase mutations – Affected 46, XY individuals have difficult to characterize external genitalia

## I. Primordial Germ Cells

1. During their migration the PGCs proliferate extensively via normal mitosis
2. Mitotic division requires DNA duplication during S-phase of the cell cycle, followed by defined stages that allow for even splitting of the duplexed chromosomes.
   a. Four main stages of mitosis: prophase, metaphase, anaphase, and telophase.

3. Gametogenesis requires the generation of HAPLOID gametes and needs two rounds of cell division without an intervening synthesis phase. This is meiosis.
4. In the testis, the process of meiosis does not begin until after birth. The PGCs remain dormant until puberty
5. In the ovaries, meiosis begins in utero and is suspended in the primordial oogonium until a follicle is selected for maturation during a menstrual cycle.
6. The stages of meiosis are similar to those in mitosis but have distinct differences. The most notable being in Prophase I.

## J. Meiosis

1. Prophase I begins after DNA synthesis and lasts much longer than mitotic prophase
   a. Leptotene stage – chromosomes condense
   b. Zygotene stage – homologous chromosomal pairs align and pair forming the synaptonemal complex

c. Pachytene – recombination occurs

d. Diplotene – synaptonemal complex breaks down and chromosomes split apart

   i. This is where the arrest occurs in the oocyte
   ii. Intense metabolic activity happens when Prophase I is restarted to prep the ovum for fertilization

e. Diakinesis – Homologous chromosomal pairs align for separation

2. Metaphase I follows Prophase I and the homologous pairs of chromosomes are split
3. Anaphase I and Telophase I proceed quickly and are followed by interkinesis/interphase II
4. In the developing sperm Meiosis II stages progress and are completed rapidly.
5. In the developing oocyte, Meiosis II begins but arrests even in the ovulated gamete, at Metaphase II

   a. Meiosis II is only completed upon fertilization

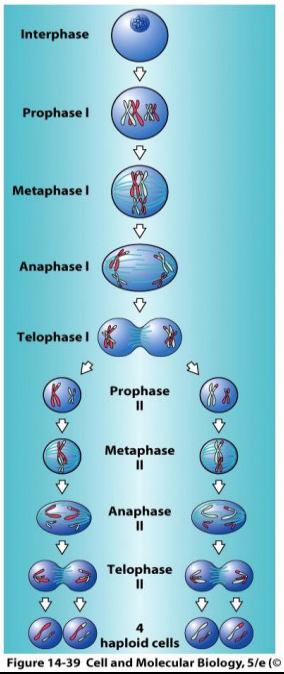


Figure 14-39 Cell and Molecular Biology, 5/e (©

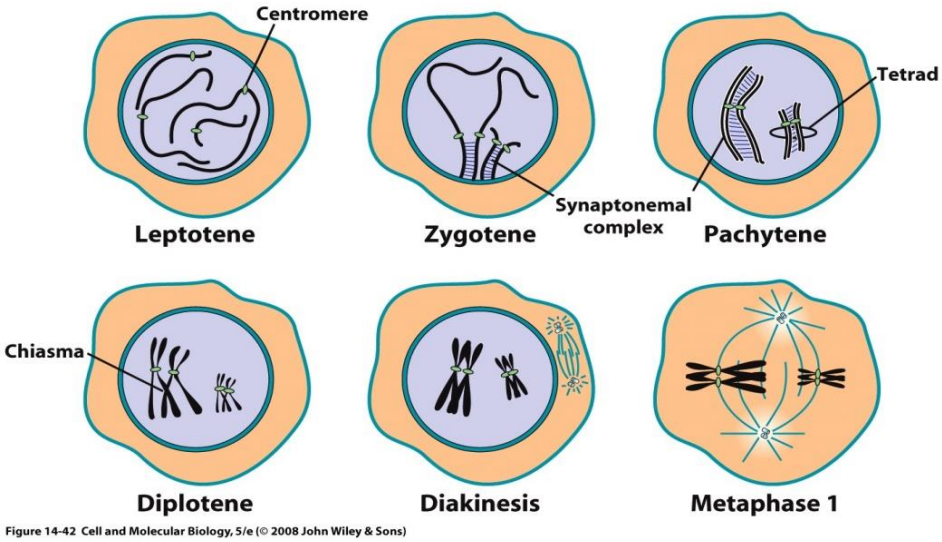



Figure 14-42 Cell and Molecular Biology, 5/e (© 2008 John Wiley & Sons)

## Answers to Study Questions

*Taken from Langman's embryology*

1. It is said the external genitalia in 46, XX and 46, XY individuals are homologous. What are the anatomic homologues, and what are their embryological origins?

The external genitalia passes through an indifferent stage during which it is impossible to differentiate between the two genotypes. Under the influence of testosterone, these structures undergo virilization, but the derivatives are homologous between biologic sexes. These homologies include (1) the clitoris and penis, derived from the genital tubercle; (2) the labia majora and scrotum, derived from the genital swellings that fuse under the control of testosterone and (3) the labia minora and penile urethra, derived from the urethral folds that also fuse under the control of testosterone. During early stages, the genital tubercle is larger 46, XX fetuses, and this has led to misidentification of biological sex by ultrasound.

2. After several years of trying to become pregnant, a young patient seeks consultation. Examination reveals she has a bicornate uterus. How might this type of structural difference have occurred?

The uterus is formed by fusion of the lower portions of the paramesonephric (Müllerian) ducts. Numerous abnormalities have been described; the most common consists of two uterine horns (bicornuate uterus). Complications of this defect include difficulties in becoming pregnant, high incidence of spontaneous abortion, and abnormal fetal presentations. In some cases, a part of the uterus has a blind end (rudimentary horn), causing problems with menstruation and abdominal pain.